DAY 9 OF 30

Data Factory pipelines

The part that runs everything.
On time, in order, with retries.

Copy | Schedule | Orchestrate

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THE PROBLEM

Good code is useless if it runs at the wrong time.

1	**A script on a laptop**	Runs when someone remembers
2	**Cron on a server**	No retries, no visibility
3	**Jobs out of order**	Reports built on old data

Someone has to decide **what runs, when, in what order, and what happens on failure**.

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THE SIMPLE VERSION

A delivery schedule for data.

A logistics planner does not drive the trucks. They decide which truck leaves when, what it waits for, and what to do if one breaks down.

A pipeline is **that planner** for your data jobs.

✓ **Activities**	The individual jobs
✓ **Dependencies**	What waits for what
✓ **Triggers**	When it all starts

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HOW IT WORKS

Building blocks of a pipeline

1. **Copy data** moves data from a source to a destination
2. **Notebook** runs your Spark code
3. **Dataflow** and **Stored procedure** run other logic
4. **ForEach** and **If** add loops and branches
5. **Triggers** start it on a schedule or on an event

Orchestrate, do not transform

Pipelines are best at moving and sequencing. Heavy logic belongs in notebooks or SQL.

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FOR ENGINEERS

Habits that save you later

Parameterise from day one

One pipeline driven by a list of tables beats fifty copies of the same pipeline.

Make every run repeatable

A rerun after failure should not create duplicates. Design loads to be idempotent.

Use event triggers where they fit

Starting when a file lands, or when another job finishes, beats polling on a schedule.

Plan the failure path

Retries, alerts and a clear owner. A silent failure is worse than a loud one.

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THE HONEST PART

Pipeline or something else?

USE A PIPELINE

Moving data between systems

Running steps in a set order

Scheduling and retrying jobs

LOOK ELSEWHERE

Complex row-level transformations

Data that arrives every second

Logic that needs unit tests

> **The catch:** pipelines grow by copy and paste. Without parameters, a small platform becomes hundreds of near-identical pipelines nobody dares to change.

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FROM REAL WORK

Pipelines on a real project

A people-analytics platform where HR data was shared as CSV files by hand.

1. **Manual CSV workflows** were replaced with pipelines
2. **Triggers** started the loads automatically
3. **Data refreshed daily** with no manual step
4. **PySpark notebooks** did the transformation work

The pipeline did not make the data better. It made it **arrive, every day**.

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SAVE THIS

Pipelines in five lines

1. **Activities** do the work
2. **Dependencies** set the order
3. **Triggers** decide when
4. **Parameters** keep it maintainable
5. **Failure handling** is part of the design

Tomorrow: Dataflows Gen2.
Cleaning data without writing code.

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